



SVP-17-019

10 CFR 50.73

March 22, 2017

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Units 1 and 2
Renewed Facility Operating License Nos. DPR-29 and DPR-30
NRC Docket Nos. 50-254 and 50-265

Subject: Licensee Event Report 254/2017-001-00, "Secondary Containment Interlock Doors
Opened Simultaneously"

Enclosed is Licensee Event Report (LER) 254/2017-001-00, "Secondary Containment Interlock
Doors Opened Simultaneously," for Quad Cities Nuclear Power Station, Unit 1.

This report is submitted in accordance with 10 CFR 50.73 (a)(2)(v)(C) which requires the reporting
of any event or condition that could have prevented the fulfillment of the safety function of structures
or systems that are needed to control the release of radioactive material.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Mr. W. J. Beck at
(309) 227-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "Scott Darin", written over a horizontal line.

Scott Darin
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

NRC FORM 366 (06-2016)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 10/31/2018				
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/) Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.											
1. FACILITY NAME Quad Cities Nuclear Power Station Unit 1					2. DOCKET NUMBER 05000254		3. PAGE 1 OF 5				
4. TITLE Secondary Containment Interlock Doors Opened Simultaneously											
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED		
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER	
01	24	2017	2017	- 001	- 00	03	22	2017	Quad Cities Nuclear Power Station Unit 2	05000265	
									FACILITY NAME	DOCKET NUMBER	
									N/A	N/A	
9. OPERATING MODE		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)									
1		☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
		☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
		☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
		☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
10. POWER LEVEL 098		☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
		☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
		☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
		☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
		☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
					☐ 50.73(a)(2)(i)(C)			☐ OTHER		Specify in Abstract below or in NRC Form 366A	
12. LICENSEE CONTACT FOR THIS LER											
LICENSEE CONTACT Mark Humphrey – Regulatory Assurance									TELEPHONE NUMBER (Include Area Code) (309) 227-2810		
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT											
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX		
B	NG	DR	N/A	Y							
14. SUPPLEMENTAL REPORT EXPECTED						15. EXPECTED SUBMISSION DATE			MONTH	DAY	YEAR
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO									N/A	N/A	N/A
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) On January 24, 2017, at 10:00 hours, Operations was notified that both doors in the secondary containment interlock on the 595 foot elevation between the Reactor Building (RB) and the Unit 2 Reactor Feed Pump room (located inside the Turbine Building (TB)) were opened simultaneously for approximately 3 seconds. The failure of this interlock caused a loss of secondary containment per Technical Specification (TS) 3.6.4.1, Condition A. The doors were immediately reclosed, and the secondary containment boundary was immediately reestablished. Operators verified the RB (secondary containment) differential pressure was maintained operable at greater than 0.10 inches of water vacuum. Secondary containment remained available and functional during the event since the secondary containment interlock was immediately restored by closing the doors and since the RB differential pressure was maintained during the event. The RB is a common volume to both Units 1 and 2, and an interlock failure can impact the secondary containment for both units. The cause of the interlock failure was due to a dirty contact that caused the interlock relay to stick. This allowed the second door to open before the first door was secured. Corrective actions included inspecting and cleaning of the interlock relay contacts. Steps will be added to the Preventative Maintenance Work Orders to perform a visual inspection and cleaning of interlock relays. The safety significance of this event was minimal. Given the impact on the secondary containment, this report is submitted (for Units 1 and 2) in accordance with the requirements of 10 CFR 50.73 (a)(2)(v)(C), which requires the reporting of any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Quad Cities Nuclear Power Station Unit 1	05000254	2017	- 001	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric - Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Both doors in the secondary containment interlock between the Reactor Building and the Unit 2 Reactor Feed Pump room were opened simultaneously and caused a momentary loss of secondary containment.

A. CONDITION PRIOR TO EVENT

Unit: 1 / 2

Event Date: January 24, 2017

Event Time: 10:00 hours

Reactor Mode: 1 / 1

Mode Name: Power Operation / Power Operation

Power Level: 98% / 100%

B. DESCRIPTION OF EVENT

On January 24, 2017 at 10:00 hours, the Operations Department was notified that both doors [DR] in the secondary containment [NG] interlock [IEL] on the 595 foot elevation between the Reactor Building (RB) [NG] and the Unit 2 Reactor Feed Pump (RFP) [SK] room (located inside the Turbine Building (TB)) [NM] were opened simultaneously for approximately 3 seconds. The failure of this interlock caused a loss of secondary containment per Technical Specification (TS) 3.6.4.1, Condition A. The doors were immediately reclosed, and the secondary containment boundary was immediately reestablished. Operators verified the RB (secondary containment) differential pressure was maintained operable at greater than 0.10 inches of water vacuum.

The Operations Field Supervisor was then dispatched to the interlock to investigate the condition. It was determined that the condition that led to both doors being open simultaneously could be recreated. At no time during the initial investigation were the doors opened at the same time. The doors were then taken out of service to allow repairs.

Secondary containment remained available and functional during the event since the secondary containment interlock was immediately restored by closing the doors and since the RB differential pressure was maintained during the event. A review of the Station Event Recorder (SER) verified that RB low differential pressure alarms were not received during this event. The RB is a common volume to both Units 1 and 2, and an interlock failure can impact the secondary containment for both units.

On January 24, 2017, at 14:00 hours, ENS #52508 was made to the NRC under 10 CFR 50.72(b)(3)(v)(C), to report this event as an event or condition that could have prevented the fulfillment of a safety function.

At the time of the event, another individual had traversed through the RB door when a contractor started transitioning through the TB door. As they opened the TB-side door the RB-side door was noticed open and in the process of closing. At no time during the event were the open door indicating lights illuminated. The door was fully closed in three seconds. It was found that the sticking relay made up the logic falsely signaling to the system that the opposite door was closed.

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NARRATIVE

Given the impact on the secondary containment, this report is submitted (for Units 1 and 2) in accordance with the requirements of 10 CFR 50.73(a)(2)(v)(C), which requires the reporting of any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material.

C. CAUSE OF EVENT

The cause was determined to be a dirty contact on an interlock relay that caused the relay to stick. The contact was inspected, cleaned, and tested satisfactory. The door was operated approximately 35 times and the sticking relay could not be repeated.

D. SAFETY ANALYSIS**System Design**

The function of the secondary containment is to contain, dilute, and hold up fission products that may leak from primary containment following a Design Basis Accident (DBA). In conjunction with operation of the Standby Gas Treatment System (SBGTS) [BH] and closure of certain valves [V] whose lines penetrate the secondary containment, the secondary containment is designed to reduce the activity level of the fission products prior to release to the environment, and to isolate and contain fission products that are released during certain operations that take place inside primary containment, when primary containment is not required to be operable, or that take place outside primary containment.

Updated Final Safety Analysis Report (UFSAR) Section 6.2.3.1 provides that the safety objective of the secondary containment system, in conjunction with other engineered safeguards and nuclear safety systems, is to limit the release of radioactive materials so that offsite doses resulting from a postulated DBA will remain below 10 CFR 100 guideline values.

The secondary containment interlocks are designed to provide personnel access to the RB from the TB while maintaining a negative differential pressure in the RB. The Unit 2 RFP interlock doors are normally secured closed by energized electromagnets. The doors are operated by a pushbutton near each door that de-energize the electromagnets and start an actuator to open the door. The doors are designed with relay logic so that only one door can be opened at a time. A red light near each door illuminates when one of the doors is open.

Safety Impact

Both Units 1 and 2 share a common RB (secondary containment). When both RFP interlock doors were opened simultaneously, this caused a momentary loss of secondary containment.

TS 3.6.4.1, Action A.1, requires restoration of secondary containment to operable status within four hours. This four hour Completion Time provides a period of time to correct the problem that is commensurate with the importance of maintaining secondary containment during Modes 1, 2, and 3, since the probability of an accident occurring during this short period where secondary containment is inoperable is minimal.

The primary purpose of the secondary containment is to minimize the ground level release of airborne radioactive materials and to provide a controlled, elevated release of the building atmosphere under accident conditions. An engineering analysis was performed to demonstrate that during the short (momentary) time that both doors of the corresponding interlock were simultaneously opened, the doors were not open sufficiently long enough to cause a RB

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low differential pressure alarm, hence no loss of secondary containment differential pressure occurred. Secondary containment would have sufficiently contained radioactive materials during a Loss of Coolant Accident (LOCA) such that all current dose limits would remain to be met. As a result, the system safety function of secondary containment would have been maintained throughout the event. Therefore, the dose consequence from postulated releases from the RB during this short duration would remain to be bounded by the existing design basis LOCA dose analysis. The safety significance of this event was minimal.

The engineering analysis that was performed demonstrated this event did not constitute a Safety System Functional Failure (SSFF). (Reference NEI 99-02, Revision 7, Regulatory Assessment Performance Indicator Guideline, Section 2.2, Mitigating Systems Cornerstone, Safety System Functional Failures, Clarifying Notes, Engineering analyses.) As such, this event will not be reported in the NRC Performance Indicator (PI) for safety system functional failures since an engineering analysis was performed which determined that the system was capable of performing its safety function during this event when both doors of the secondary containment interlock were momentarily simultaneously opened.

Risk Insights

The plant Probabilistic Risk Assessment (PRA) model gives no credit to the secondary containment and does not include it in the model, hence the as-found conditions did not contribute to an increase in risk. In addition, the physical integrity of the secondary containment structure was never compromised and the primary containment function was never lost.

Although secondary containment was momentarily inoperable per TS 3.6.4.1, Condition A, when the Unit 2 RFP interlock doors were opened simultaneously due to an intermittently binding interlock relay contact, there was no DBA condition in progress, and secondary containment function was restored immediately when one of the doors was closed. RB differential pressure was maintained during the event.

In conclusion, the overall safety significance and impact on risk of this event were minimal.

E. CORRECTIVE ACTIONS**Immediate:**

1. The dirty contact was inspected, cleaned and tested satisfactorily. The door was operated approximately 35 times and the issue did not reoccur.

Follow-up:

1. Add steps to all interlock Preventive Maintenance and current Work Orders for visual inspection, voltage checks, and cleaning of all interlock relays.

F. PREVIOUS OCCURRENCES

The station events database, LERs, and INPO Consolidated Event System (ICES) were reviewed for similar events at Quad Cities Nuclear Power Station. This event was a failure of a secondary containment interlock door caused by a dirty contact on an interlock relay, which allowed two interlock doors to be opened simultaneously. Based on the conditions of this event, causes, and associated corrective actions, the events listed below, although similar in topic, are not considered significant station experiences that would have directly contributed to preventing this event.

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NARRATIVE

- LER 254/2014-002-00, 06/02/14, Reactor Building Interlock Doors Opened Simultaneously Cause Loss of Secondary Containment (04/1/14) - Both doors in the secondary containment interlock on the 595 foot elevation from the Unit 2 Reactor Feed Pump room to the Reactor Building were opened simultaneously. The cause of the interlock failure was due to a malfunctioning interlock door hydraulic actuator and time delay relays had allowed the second door to open before the first door was secured. Corrective actions included replacing the failed actuator, adjusting the limit switch, and a set point change to resolve relay time delay issues. Since this event cause is different from the current event cause (sticking relay), the previous interlock failure event is not directly applicable to the event of this current LER.
- LER 254/2015-006-00, 6/24/15, Interlock Doors Opened Simultaneously Cause Loss of Secondary Containment (04/28/15) - the control room was notified that the Unit 1/2 emergency diesel generator (EDG) room manual side door and the Unit 1 reactor building (RB) side door, as a part of the personnel interlock to the 1/2 EDG room, were opened simultaneously. The failure of this interlock caused a temporary loss of secondary containment (inoperable) per Technical Specification (TS) 3.6.4.1, Condition A. The 1/2 EDG manual interlock door was closed immediately, and the secondary containment boundary was immediately reestablished. Operators verified the RB (secondary containment) differential pressure remained negative during this event. It was determined that the latch did not engage into the strike plate, hence this failure mechanism is not directly applicable to the event of this current LER.

G. COMPONENT FAILURE DATA

Failed Equipment: Relay
Component Manufacturer: General Electric
Component Model Number: CR122
Component Part Number: N/A

This event has been reported to ICES as Report No. 326170